

Team Composition, Recomposition, and Role Selection in Long-Term Educational Programs: A Theoretical Framework Using Multiple Criteria Approach

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Abstract. The topic of team composition, recomposition, and role selection in long-term educational programs that comprise of multiple smaller courses, such as coding bootcamps, has received limited attention in the literature, despite its importance. However, the rise of large-scale online learning environments such as MOOCs has highlighted the need for effective strategies to address the high drop-out rates and low quality graduates observed in these contexts. In this paper, we propose to build a theoretical framework by using a multiple-criteria technique that incorporates personalities, agile, and gamification to address this problem. This framework will be used to better understand the dynamics of team composition, recomposition, and role selection in virtual teams within such educational programs.

Keywords. Team composition, Team recomposition, Role selection, Long-term educational programs, MOOCs, Team performance

Introduction

Massive Open Online Courses (MOOCs) have revolutionized education, providing large-scale, accessible, and affordable education to learners without geographical or financial constraints [1]. Team-based learning and collaborative projects are often employed in long-term educational programs, such as coding bootcamps on MOOCs. Effective team composition, recomposition, and role selection are crucial to the success of these educational programs. Research has indicated that team composition can significantly impact team performance, as diverse teams are often better equipped to solve complex problems and achieve their goals [2]. Furthermore, team composition and recomposition are not static processes, as teams may need to be reconfigured or new members added to address changing needs or circumstances. Therefore, a comprehensive understanding of the factors influencing team composition and effective team recomposition is necessary to ensure the success of long-term educational programs.

Furthermore, effective role selection is crucial, as individuals may have different skills, strengths, and learning styles that can affect their performance and contribution to

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the team. Failing to consider these factors can lead to unproductive and dysfunctional teams that may impact individual students' learning outcomes and experience, as well as the overall success of the educational program.

For example, in a study of a MOOC on programming, students who worked in teams with diverse skill sets and backgrounds were found to have better learning outcomes than those who worked in teams with homogeneous skill sets [3].

On the other hand, the wrong team composition can lead to frustration and disengagement, with some students feeling unsupported or undervalued in their team. This can result in lower motivation, poor performance, and even dropouts [4]. Therefore, the topic of team composition, recomposition, and role selection in long-term educational programs is essential because it can impact the success of educational programs and the learning outcomes of individual students. Addressing this topic can lead to more effective team-based learning and collaborative projects, enhancing student learning and preparing students for career success.

This paper proposes a theoretical framework for team composition, recomposition, and role selection in long-term educational programs. The framework uses a multiple-criteria approach that incorporates personalities, agile techniques, and gamification to address the challenges of team-based learning and collaboration. The methodology used in this paper includes a literature review, data analytics, and a multiple-criteria approach questionnaire to gather data and analyze the effectiveness of the proposed framework. The framework proposed in this paper can help instructors and program administrators optimize team composition, improve team dynamics, and enhance team-based learning and collaboration in long-term educational programs [3,4].

1. Literature

1.1. *Team composition in educational programs*

Previous research has shown that team composition is a critical factor in the success of educational programs [2]. The composition of a team can impact various aspects of learning outcomes, including academic performance, engagement, and satisfaction. Factors influencing team composition include ability level, personality traits such as DISC or MBTi, individual accountability, communication skills, demographics, and passion for the subject. Considering these criteria, building more effective and productive teams in educational programs is possible.

The impact of team composition on learning outcomes has been explored in various studies. For example, a study of a software engineering course found that students who could choose their teams based on shared interests and skills had higher levels of satisfaction and performance than those assigned to teams [6]. Similarly, a study of a MOOC on programming found that students who worked in teams with diverse skill sets and backgrounds had better learning outcomes than those who worked in teams with homogeneous skill sets [7].

1.2. *Recomposition and role selection in educational programs*

Next, it is essential to discuss previous research on recomposition and role selection in educational programs and their relevance in the context of long-term educational programs such as software engineering MOOCs. Recomposition refers to the process of

changing team membership during a long-term educational program, while role selection refers to assigning team members specific roles within the team project. These processes are important for optimizing team dynamics and learning outcomes in long-term educational programs. Previous research has examined factors impacting recomposition and role selection, including ability level, personality, communication skills, and team performance. For instance, a study by Aggarwal et al. [7] found that considering students' cognitive styles can improve the effectiveness of team recomposition in educational programs. Similarly, a study by Özgen et al. [8] found that effective role selection in project-based learning can improve student motivation and engagement. Previous research suggests that careful attention to recomposition and role selection is essential for optimizing team dynamics and learning outcomes in long-term educational programs, including software engineering MOOCs.

1.3. *Multiple criteria approach in team composition*

The multiple criteria approach has been widely used in team composition research to optimize team dynamics and achieve better learning outcomes in long-term educational programs such as software engineering MOOCs. This approach involves considering a variety of criteria when building and reconfiguring teams, such as ability level, personality, individual accountability, communication skills, demographics, and passion. Several algorithms can be used in the multiple criteria approach, such as Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), Analytic Hierarchy Process (AHP), Simple Additive Weighting (SAW), Weighted Product (WP), and Data Envelopment Analysis (DEA). Previous research has shown that using the multiple criteria approach can improve team performance and student satisfaction. For example, a study by Dabbagh and Kitsantas [9] found that considering personality traits and communication skills when building teams in an online learning environment can improve team cohesion and student satisfaction.

Similarly, a study by Daim et al. [10] found that the multiple criteria approach can effectively predict team performance in virtual teams. Another study by Sandhya and R. Garg [11] used the AHP algorithm to identify the most critical criteria for team composition in a software engineering project and found that ability level, personality, and communication skills were the most critical factors. Overall, previous research highlights the importance of considering multiple criteria and using appropriate algorithms when building and reconfiguring teams in long-term educational programs and the potential benefits of the multiple criteria approach for optimizing team dynamics and achieving better learning outcomes.

1.4. *Personality, agile, and gamification in educational programs*

Previous research has identified the importance of personality, agile techniques such as Scrum, and gamification elements such as points, badges, and leaderboards in improving educational program team dynamics and learning outcomes. Personality is a key factor in team composition, and matching individuals based on their personality traits can lead to better collaboration and communication. Research by Kraiger and Kirkpatrick [12] found that personality matching based on the MBTI and DISC assessment tools can improve team cohesion and reduce conflicts in virtual teams.

Agile techniques such as Scrum have also improved team dynamics and productivity. Scrum is a framework that emphasizes team collaboration and

communication and involves regular meetings such as daily stand-ups and sprint reviews to track progress and identify areas for improvement. Ramin et al. [13] found that using Scrum in a software engineering project improved team collaboration, communication, and productivity. Scrum can be used in educational programs to encourage team collaboration and accountability and to help instructors monitor progress and identify areas for improvement.

Gamification elements such as points, badges, and leaderboards can also improve learning outcomes and motivation in educational programs. Points can be used to track progress and provide student feedback, while badges can be used to reward achievement and provide a sense of accomplishment. Leaderboards can be used to create a sense of competition and encourage students to work harder to improve their ranking. Research by [14] found that incorporating gamification elements can improve engagement and motivation in online learning environments. In educational programs, gamification elements can be used to increase motivation and engagement and provide students with a sense of achievement and progress. By incorporating these elements into team activities, instructors can encourage collaboration and competition and help students stay engaged and motivated throughout the program.

In conclusion, the literature review highlights the importance of team composition, recomposition, and role selection in long-term educational programs such as software engineering MOOCs. The criteria examined in this literature review include ability level, personality, individual accountability, communication skills, demographics, and passion. The multiple criteria approach has been identified as an effective way to optimize team dynamics and achieve better learning outcomes. Using appropriate algorithms, such as TOPSIS, AHP, SAW, WP, and DEA, can help instructors build and reconfigure more effective and productive teams and achieve better learning outcomes. By carefully considering sub-criteria such as academic background, MBTI and DISC matching, pre-camp behavior, available time, and nationality, instructors can ensure that team members are well-matched and can work together effectively. The literature review also highlights the need for further research in this area, particularly with regard to the effectiveness of different algorithms and the impact of team composition on specific learning outcomes. This literature review provides a foundation for future research on team composition, recomposition, and role selection in long-term educational programs such as software engineering MOOCs.

2. Theoretical Framework

The theoretical framework proposed in this paper integrates multiple criteria to examine team composition, recomposition, and role selection in long-term educational programs, such as coding bootcamps and MOOC-based programs with team projects. This comprehensive approach addresses the complexities and challenges associated with virtual team dynamics and improves learning outcomes, engagement, and overall team performance.

2.1. Overview of the multiple-criteria approach

The multiple-criteria approach provides a structured method for evaluating and prioritizing factors that may influence team composition, recomposition, and role selection [15]. This approach is advantageous in educational contexts, where diverse

factors, such as ability level, personality, individual accountability, communication skills, demographic, and passion, can significantly impact team dynamics and learning outcomes [16,17]. By systematically considering these factors, the multiple-criteria approach offers a more comprehensive understanding of the interplay between different aspects of team formation and their effects on team performance.

2.2. *Criteria for team composition*

Team composition is a critical factor in the success of long-term educational programs such as coding bootcamps and MOOCs. In order to achieve optimal learning outcomes and high levels of engagement, it is essential to carefully consider the abilities, personalities, and backgrounds of team members. To this end, we propose a set of six criteria for team composition: ability level, personality, individual accountability, communication skills, demographics, and passion. Ability level includes subcriteria such as an academic background in software engineering, degree of education, work experience background, pre-camp (some courses that learners need to study before entering bootcamp), behavior, skill contribution to the project, and related courses studied previously. Personality includes subcriteria such as MBTI and DISC matching with the subject (some personalities like to study in some subjects), game style (preferences of gamification elements), learning style survey (survey of the learning styles that learners fill in), learning style pre-camp (data analytics result of learning styles of users in pre-camp), and grit. Individual accountability includes subcriteria such as pre-camp behavior (data analytics result of individual accountability of users in pre-camp), available time, and motivation. Communication skills include subcriteria such as pre-camp behavior (data analytics result of communication skills of users in pre-camp) and past collaboration such as work experience. Demographics include subcriteria such as gender, nationality, and location. Passion includes subcriteria such as harmonious passion, obsessive passion, passion for the pre-camp survey subject (survey of the passion of learners toward each subject), and passion for the pre-camp survey role (survey of the passion of learners toward each scrum role). By using these criteria and subcriteria, educational programs can create more diverse and influential teams that are better equipped to handle complex projects and achieve success.

2.3. *Criteria for role selection*

The Agile methodology has gained popularity in software engineering education in recent years. The Agile approach emphasizes collaboration, flexibility, and rapid iteration, making it an attractive option for educational programs focused on project-based learning. However, effective implementation of Agile techniques requires careful consideration of team composition and role selection. One approach to addressing these issues is the Analytic Hierarchy Process (AHP), a widely used decision-making tool that can help evaluate and prioritize multiple criteria in complex decision-making scenarios. Several studies have utilized AHP in role selection, such as selecting key personnel in construction projects [18] and selecting suppliers in the automotive industry [19]. By using AHP to select roles in Agile teams, instructors can help ensure that team members are well-suited to their roles, leading to more effective collaboration and better project outcomes. In this paper, we propose using AHP to select the roles of Product Owner and Scrum Master in an educational context, based on criteria and sub-criteria related to ability level, personality, individual accountability, communication skills, demographic

factors, and passion toward the course. Based on the six criteria and their respective sub-criteria, the following criteria should be used for the selection of roles in the Scrum process:

For the Product Owner role, the criteria of "Academic Background" with sub-criteria of "Degree of Education" and "Work Experience Background," as well as the criteria of "Passion" with sub-criteria of "Harmonious Passion" and "Passion in Pre-camp Survey Subject" should be considered. This is because the Product Owner is responsible for setting the project's vision and direction; having a strong educational and work background and a passion for the subject matter can contribute to better decision-making and effective leadership.

For the Scrum Master role, the criteria of "Individual Accountability" with sub-criteria of "Available Time" and "Motivation," as well as the criteria of "Communication Skills" with sub-criteria of "Amount of Past Collaboration" and "Pre-Camp Behavior" should be considered. This is because the Scrum Master is responsible for ensuring the team follows the Scrum process and facilitates communication and collaboration among team members. Thus, having strong communication skills and a high level of individual accountability can contribute to effective leadership in this role

2.4. *Team Recomposition and Gamification for Team Recomposition*

Team composition is a critical factor in the success of long-term educational programs, and the need for recomposition arises when existing teams are not functioning effectively. To address this problem, a two-phase recomposition system is proposed. The first phase determines the need for recomposition based on the established criteria of ability level, personality, individual accountability, communication skills, demographics, and passion toward the course. The second phase is to decompose some teams and compose new ones based on the same criteria. This section will discuss the proposed process and gamification system for team recomposition in educational programs.

In the first phase, data from surveys and learning management systems will be analyzed to evaluate the performance of existing teams based on the established criteria. The engine will identify the need for recomposition if the team is not functioning effectively. In the second phase, the gamification system will be applied to assign roles in the new team. Students with high scores or more badges in specific criteria will have a higher chance of becoming Scrum Masters or Product Owners. The proposed gamification system will increase students' motivation and engagement in the learning process and encourage them to perform well in the established criteria.

The below game elements will be used to determine the roles of Scrum Master and Product Owner.

For the Scrum Master role, potential gamification elements could include:

1. Leadership Score: This score would be based on the individual's ability to lead and facilitate group discussions and meetings.
2. Problem-solving ability: This could be evaluated through the completion of problem-solving challenges within the learning management system.
3. Time Management: This could be evaluated by the individual's ability to complete tasks and meet deadlines within the learning management system.

For the Product Owner role, potential gamification elements could include:

1. Subject Matter Expertise: This could be evaluated through the completion of quizzes and assessments related to the course subject matter.

2. Creativity: This could be evaluated by completing creative tasks and projects within the learning management system.
3. Strategic Thinking: This could be evaluated by the individual's ability to prioritize tasks and make decisions related to the project.

The gamification elements for both roles should be tailored to the specific educational program and its objectives. The weights assigned to each gamification element in the AHP process should also be determined based on the program's specific needs and goals.

We propose using these game elements as sub-criteria of the current criteria. The new sub-criteria are leadership score, problem-solving ability, time management, subject matter expertise, creativity, and strategic thinking. To incorporate these sub-criteria, we suggest adding them to the existing criteria based on their relevance. Specifically, leadership score and problem-solving ability could be added to the ability level criterion, time management could be added to individual accountability, subject matter expertise and creativity could be added to passion, and strategic thinking could be added to the communication skills criterion.

Adding these sub-criteria will help provide a more comprehensive and detailed evaluation of team composition and role selection. By considering a more comprehensive range of skills and abilities, the proposed framework will help ensure that teams are composed of individuals with the necessary expertise and skills to complete the project successfully. Additionally, incorporating this sub-criteria will help to identify individuals with strong leadership skills and problem-solving abilities, which are essential qualities for effective team collaboration and project success. Overall, the addition of these sub-criteria will improve the accuracy and effectiveness of the team composition and role selection process, leading to better outcomes for educational programs and project-based learning.

2.5. *Grouping by heterogeneity*

Team composition is a critical factor in the success of collaborative learning projects, and research has shown that heterogeneous teams tend to outperform homogeneous teams in terms of creativity, problem-solving, and overall performance [20]. To ensure maximum heterogeneity in team composition, we propose using heterogeneity indices, such as the Entropy Index and the Gini Index, combined with the Analytic Hierarchy Process (AHP) and a genetic algorithm. The AHP allows us to systematically evaluate multiple criteria and sub-criteria related to team composition, while the genetic algorithm helps to optimize the heterogeneity index and create diverse and influential teams. By using these techniques in combination, we can create teams that are both well-suited to their respective tasks and highly diverse in terms of skill sets, backgrounds, and perspectives. This can lead to more creative problem-solving, better communication, and, ultimately, better learning outcomes for all team members.

2.6. *Our Proposed framework*

The team composition, role selection, and team recomposition process involved multiple steps. Firstly, we utilized the Analytic Hierarchy Process (AHP) to calculate scores for each criterion identified in the methodology section. Secondly, we grouped the students with the heterogeneity index and genetic algorithm in conjunction with AHP scores. The heterogeneity index helped ensure the teams were diverse and well-rounded, leading to

better learning outcomes and project success. The genetic algorithm further optimized the team composition to ensure the best match between students' abilities and preferences. Thirdly, we utilized AHP again to select the Scrum Master and Product Owner from within the groups. This step helped ensure that team members were assigned roles based on their strengths and abilities, leading to better collaboration and project outcomes. Finally, after two weeks of working together, we employed the team recomposition process, using gamification elements to motivate students to participate. This step aimed to optimize the team composition further and address any issues during the project's initial phase. Overall, this process helped us to identify the optimal team composition and role selection, leading to better learning outcomes and project success.

3. Research Methodology

This study employs a mixed-methods research design to explore the effectiveness of a proposed framework for team composition, recomposition, and role selection in long-term educational programs. The study involves the following steps:

1. Setting up the low-code coding bootcamp, which requires students to finish one month full-time and three months part-time. The bootcamp will have six subjects: computer science fundamentals, HTML, JavaScript, database, business flow, and low-code programming.
2. Conducting a pre-camp study where students study the computer science fundamentals subject alone without a group for one month. The next two months will be spent studying the rest of the subjects in groups. All of these subjects will use a learning management system for online learning.
3. Opening registration for the bootcamp, targeting 1000 students.
4. Administering a survey to students before the camp to gather input for sub-criteria, with some input taken automatically from the data analytics in the learning management system.
5. Focusing on the values of personality and passion toward the course. The value of this criteria will differ for each course.
6. Separating students into two experimental groups equally: a control group, which will use randomly formed teams, and an experimental group, which will use the proposed method for grouping and recomposition.
7. Assigning four people per group.
8. Applying the proposed framework for grouping and role selection.
9. Assigning three roles according to the scrum process: team member, scrum master, and product owner for the learning group.
10. Using Scrum for learning context such as everyone needs to set up learning goals for each sprint, the scrum master will manage everyone to learn within the plan, and the product owner will set up which subject each group needs to study.
11. Dividing each subject into two phases: study phase and project phase. In the study phase, Scrum is used for learning, and in the project phase, Scrum is used for project management.
12. Evaluating the need for recomposition after two weeks based on the criteria for recomposition. Role selection will be applied if necessary.

13. Evaluating the results of the bootcamp using qualitative criteria such as quality of code, participation, team work, and quantitative criteria such as the number of graduated students, grades, and quiz scores.

Data will be collected from a variety of sources, including surveys, learning management system analytics, and student project work. Data analysis will involve both quantitative and qualitative methods. Quantitative data will be analyzed using statistical software such as SPSS to examine the differences in learning outcomes between the control and experimental groups. Qualitative data will be analyzed using thematic analysis to identify common themes in the data related to team composition, recomposition, and role selection. Potential limitations of the study include the potential for self-selection bias among participants, the limited duration of the bootcamp, and the potential for extraneous variables to affect learning outcomes. To address these limitations, efforts will be made to recruit a diverse range of participants and to carefully control for potential confounding variables in data analysis.

This study aims to provide valuable insights into the effectiveness of a multiple criteria approach to team composition, recomposition, and role selection in long-term educational programs.

4. Conclusion

In conclusion, team composition, recomposition, and role selection in long-term educational programs is a complex and vital issues that can significantly impact learning outcomes and student engagement. Through the use of a multiple criteria approach that incorporates personality, agile techniques, and gamification, this paper proposes a theoretical framework that can help to better understand the dynamics of team composition and role selection in virtual teams within educational programs. The proposed framework offers a structured and systematic way of evaluating potential team members based on their ability level, personality, individual accountability, communication skills, demographic factors, and passion toward the subject. The framework can also be customized to fit the specific needs of different educational programs and can be used to inform decisions related to team composition, recomposition, and role selection. By using this framework, educators and program administrators can improve the quality of educational programs and better support the learning and development of their students. Future research can continue to explore the effectiveness and applicability of this framework in different educational contexts.

References

- [1] C. Sandeen, Integrating MOOCs into traditional higher education: The emerging 'MOOC 3.0' era, *Change: The magazine of higher learning*, 2013, Vol. 45, no. 6, pp. 34-39.
- [2] S. E. Page, *The Difference: How the Power of Diversity Creates Better Groups, Firms, Schools, and Societies*, Princeton University Press, 2007.
- [3] N. Durmic, Factors influencing project success: A qualitative research, *TEM Journal*, 2020, Vol. 9, no. 3, pp. 1011-1020.

- [4] T. Staubitz and C. Meinel, Graded team assignments in MOOCs: Effects of team composition and further factors on team dropout rates and performance, *Proceedings of the Sixth (2019) ACM Conference on Learning@Scale*, June 2019, pp. 1-10.
- [5] M.N. Gómez and S.T. Acuña, A replicated quasi-experimental study on the influence of personality and team climate in software development, *Empirical Software Engineering*, 2014, Vol. 19, pp. 343-377.
- [6] T. Staubitz, T. Pfeiffer, J. Renz, C. Willems, and C. Meinel, Collaborative learning in a MOOC environment, *ICERI2015 Proceedings*, 2015, pp. 8237-8246.
- [7] I. Aggarwal and A.W. Woolley, 'Do you see what I see? The effect of members' cognitive styles on team processes and errors in task execution, *Organizational Behavior and Human Decision Processes*, 2013, vol. 122, no. 1, pp. 92-99.
- [8] S. Özgen, J. Alabart, and M. Medir, A team leader selection process for project based learning experiences, *2008 Annual Conference & Exposition*, June 2008, pp. 13-125.
- [9] N. Dabbagh and A. Kitsantas, Personal learning environments, social media, and self-regulated learning: A natural formula for connecting formal and informal learning, *Internet High. Educ.*, 2012, Vol. 15, No. 1, pp. 3-8.
- [10] T. U. Daim, A. Ha, S. Reutiman, B. Hughes, U. Pathak, W. Bynum, and A. Bhatla, Exploring the communication breakdown in global virtual teams, *International Journal of Project Management*, 2012, Vol. 30, no. 2, pp. 199-212.
- [11] S. Sandhya and R. Garg, Implementation of multi-criteria decision making approach for the team leader selection in IT sector, *Journal of Project Management*, 2016, Vol. 1, no. 2, pp. 67-75.
- [12] K. Kraiger and S. Kirkpatrick, An empirical evaluation of three popular training programs to improve interpersonal skills, *Journal of Psychological Issues in Organizational Culture*, 2010, vol. 1, no. 1, pp. 60-73.
- [13] F. Ramin, C. Matthies, and R. Teusner, More than code: Contributions in Scrum software engineering teams, *Proceedings of the IEEE/ACM 42nd International Conference on Software Engineering Workshops*, 2020, pp. 137-140.
- [14] J. Hamari, J. Koivisto, and H. Sarsa, Does gamification work? -- A literature review of empirical studies on gamification, *49th Hawaii International Conference on System Sciences (HICSS)*, Koloa, HI, USA, 5-8 Jan. 2016.
- [15] B. Roy, The outranking approach and the foundations of electre methods, *Theory Decis.*, 1991, Vol. 31, no. 1, pp. 49-73.
- [16] L. Patricio and M. Franco, A systematic literature review about team diversity and team performance: Future lines of investigation, *Administrative Sciences*, 2022, Vol. 12, no. 1, 31.
- [17] K.S. Hone and G.R. El Said, Exploring the factors affecting MOOC retention: A survey study, *Comput. Educ.*, 2016, Vol. 98, pp. 157-168.
- [18] C. Gadafa, P.K. Gangwar, V.S. Rawat, S.B. Dugad, L. Kumar, Multi Criteria Decision-Making Model And Approach For The Selection Of Contractor In Construction Projects, *Solid State Technology*, 2020, Vol. 63, No. 6, pp. 10571-10585.
- [19] U. Merschmann and U. W. Thonemann, Supply chain flexibility, uncertainty and firm performance: An empirical analysis of German manufacturing firms, *International Journal of Production Economics*, 2011, Vol. 130, No. 1, pp. 43-53.
- [20] A. Lockett, N. Moray, R.S. Jegers, I. Vanaelst and B. Clarysse, Entrepreneurial team development in academic spinouts: An examination of team heterogeneity, *Entrepreneurship Theory and Practice*, 2006, Vol. 30, no. 2, pp. 249-271.